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Paper Practice 9.1

Exponential Functions

WORKED KEY – Show all of your work in the space provided. Round money to the nearest dollar unless told otherwise.

1. Evaluate each function at the given value. (Warm-Up Review)

a. $f(x) = 4^x$, when $x = 0$ ANSWER: 1

b. $f(x) = 3(2)^x$, when $x = 3$ ANSWER: $3 \cdot 8 = 24$

c. $f(x) = 5(½)^x$, when $x = 2$ ANSWER: $5 \cdot ¼ = 5/4 = 1.25$

2. Determine whether each function shows exponential growth or decay. (Growth or Decay)

a. $y = 2(5)^x$ ANSWER: Growth – $b = 5 > 1$.

b. $y = 7(0.3)^x$ ANSWER: Decay – $b = 0.3 < 1$.

c. $y = 100(1.08)^x$ ANSWER: Growth – $b = 1.08 > 1$.

3. Identify a and b for each exponential function $y = a(b)^x$. (Identify a and b)

a. $y = 6(3)^x$ ANSWER: $a = 6$, $b = 3$

b. $y = 0.5(¼)^x$ ANSWER: $a = 0.5$, $b = ¼$

4. For $f(x) = 3(2)^x$, complete the table for $x = -1, 0, 1, 2, 3$. (Table) ANSWER: $f(-1) = 1.5$, $f(0) = 3$, $f(1) = 6$, $f(2) = 12$, $f(3) = 24$

5. For $y = 5(2)^x$, state the domain, the range, the y-intercept, and the equation of the asymptote. (Key Features) ANSWER: Domain: all real numbers. Range: $y > 0$. y-intercept: (0, 5). Asymptote: $y = 0$.

6. Determine whether the data shows exponential behaviour. Explain. (Exponential Behaviour)

a. x: 0, 1, 2, 3 y: 4, 12, 36, 108
ANSWER: Yes – each y is 3 times the one before it, a constant ratio.

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7. Determine whether the data shows exponential behaviour. Explain. (Exponential Behaviour)

a. x: 0, 1, 2, 3 y: 5, 9, 13, 17
ANSWER: No – the differences are a constant 4, so it is linear, not exponential.

8. Each time you fold a piece of paper in half, the number of layers doubles. Start with 1 layer. (Apply)

a. Write a function for the number of layers after x folds. ANSWER: $f(x) = 2^x$

b. How many layers after 7 folds?
ANSWER: $f(7) = 128$ layers

9. A car worth $24,000 loses 15% of its value each year. (Apply)

a. Write an exponential function for its value after x years. ANSWER: $V(x) = 24000(0.85)^x$

b. Find its value after 3 years. ANSWER: $24000(0.85)^3 = 24000(0.614125) = \$14,739$

10. A colony of 200 bacteria triples every hour. (Apply)

a. Write a function for the population after x hours. ANSWER: $P(x) = 200(3)^x$

b. How many bacteria after 4 hours?
ANSWER: $200(81) = 16,200$ bacteria

11. Error Analysis – Gus says $y = 3(0.5)^x$ is exponential growth because 3 is positive. Identify his mistake. (Error Analysis)
ANSWER: Growth or decay is decided by b, not a. Here $b = 0.5 < 1$, so it is decay.

12. Explain what an asymptote is, and why the graph of $y = a(b)^x$ never touches the x-axis when $a \neq 0$. (Explain) ANSWER: An asymptote is a line the graph approaches but never reaches. b^x is never 0, so $a(b)^x$ is never 0 and the graph never meets $y = 0$.